```
from google.colab import drive
drive.mount('/content/drive')

```

```
Mounted at /content/drive
```

```
from google.colab import drive
drive.mount('/content/drive')

```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

```
import os

os.listdir('/content/drive/MyDrive')

```

```
['Colab Notebooks',
 'Sri Ram',
 'intern_project',
 'IMG_20220903_130224.jpg',
 'IMG20220903125529.jpg',
 'IMG_20220927_184841.jpg',
 'MatLab.pdf',
 'Conference certificate.pdf',
 'Matlab 4 (1).pdf',
 'Matlab 4.pdf',
 'Matlab 2 (2).pdf',
 'Matlab 2 (1).pdf',
 'Matlab 2.pdf',
 'Matlab 3 (2).pdf',
 'Matlab 3 (1).pdf',
 'Matlab 3.pdf',
 'MatLab 5 (1).pdf',
 'MatLab 5.pdf',
 'MatLab 1 (1).pdf',
 'MatLab 1.pdf',
 'Matlab 6 (1).pdf',
 'Matlab 6.pdf',
 'Matlab 7 (1).pdf',
 'Matlab 7.pdf',
 'Matlab 8 (1).pdf',
 'Matlab 8.pdf',
 'Matlab 9.pdf',
 'MatLab10.pdf',
 'Coursera 1 (2).pdf',
 'Coursera 1 (1).pdf',
 'Coursera 1.pdf',
 'Coursera 2 (2).pdf',
 'Coursera 2 (1).pdf',
 'Coursera 2.pdf',
 'Coursera  3 (1).pdf',
 'Coursera  3.pdf',
 'Coursera 4 (2).pdf',
 'Coursera 4 (1).pdf',
 'Coursera 4.pdf',
 'Coursera 5 (2).pdf',
 'Coursera 5 (1).pdf',
 'Coursera 5.pdf',
 'Coursera  6 (1).pdf',
 'Coursera  6.pdf',
 'Coursera  7 (1).pdf',
 'Coursera  7.pdf',
 'Coursera 8 (2).pdf',
 'Coursera 8 (1).pdf',
 'Coursera 8.pdf',
 'Coursera  9 (2).pdf',
 'Coursera  9 (1).pdf',
 'Coursera  9.pdf',
 'Coursera 10 (1).pdf',
 'Coursera 10.pdf',
 'Coursera 11.pdf',
 'Coursera 12.pdf',
 'Coursera 13.pdf',
 'Data Analysis and Decision Making - I.pdf',
```

```
os.listdir('/content/drive/MyDrive/intern_project')

```

```
['genre.csv',
 'customer.csv',
 'invoice_line.csv',
 'media_type.csv',
 'playlist.csv',
 'invoice.csv',
 'playlist_track.csv',
 'employee.csv',
 'artist.csv',
 'album2.csv',
 'album.csv',
 'track.csv']
```

```
folder_path = "/content/drive/MyDrive/intern_project"

```

```
import pandas as pd
import sqlite3
import glob

conn = sqlite3.connect("musicdb.sqlite")

folder_path = "/content/drive/MyDrive/intern_project"

csv_files = glob.glob(folder_path + "*.csv")

for file in csv_files:
    table_name = file.split("/")[-1].replace(".csv", "")
    df = pd.read_csv(file)
    df.to_sql(table_name, conn, if_exists='replace', index=False)
    print("Imported:", table_name)

conn.close()
print("All CSV files imported successfully!")

```

```
All CSV files imported successfully!
```

```
import sqlite3
import pandas as pd

conn = sqlite3.connect("musicdb.sqlite")

tables = pd.read_sql("SELECT name FROM sqlite_master WHERE type='table';", conn)
tables

```

name

```
tables = pd.read_sql("SELECT name FROM sqlite_master WHERE type='table';", conn)
tables

```

name

```
from google.colab import drive
drive.mount('/content/drive')

```

```
Mounted at /content/drive
```

```
import os
os.listdir('/content/drive/MyDrive/intern_project')

```

```
['album.csv',
 'artist.csv',
 'album2.csv',
 'customer.csv',
 'invoice_line.csv',
 'employee.csv',
 'genre.csv',
 'playlist_track.csv',
 'invoice.csv',
```

```
 'media_type.csv',
 'playlist.csv',
 'track.csv']
```

```
import pandas as pd
import sqlite3
import glob

folder_path = "/content/drive/MyDrive/intern_project/"

conn = sqlite3.connect("musicdb.sqlite")

csv_files = glob.glob(folder_path + "*.csv")

print("Found CSV files:", csv_files)

for file in csv_files:
    table_name = file.split("/")[-1].replace(".csv", "")
    df = pd.read_csv(file)
    df.to_sql(table_name, conn, if_exists='replace', index=False)
    print("Imported:", table_name)

conn.close()
print("\nAll CSV files imported successfully!")

```

```
Found CSV files: ['/content/drive/MyDrive/intern_project/album.csv', '/content/drive/MyDrive/intern_project/artist.csv', '/conte
Imported: album
Imported: artist
Imported: album2
Imported: customer
Imported: invoice_line
Imported: employee
Imported: genre
Imported: playlist_track
Imported: invoice
Imported: media_type
Imported: playlist
Imported: track

All CSV files imported successfully!
```

```
conn = sqlite3.connect("musicdb.sqlite")
tables = pd.read_sql("SELECT name FROM sqlite_master WHERE type='table';", conn)
tables

```

	name
0	album
1	artist
2	album2
3	customer
4	invoice_line
5	employee
6	genre
7	playlist_track
8	invoice
9	media_type
10	playlist
11	track

```
df = pd.read_sql("SELECT * FROM customer LIMIT 5;", conn)
df

```

	customer_id	first_name	last_name	company	address	city	state	country	postal_code	phone	fax	
0	1	Luís	Gonçalves	Embraer - Empresa Brasileira de Aeronáutica S.A.	Av. Brigadeiro Faria Lima, 2170	São José dos Campos	SP	Brazil	12227-000	+55 (12) 3923-5555	+55 (12) 3923-5566	luisg@em
1	2	Leonie	Köhler	None	Theodor-Heuss-Straße 34	Stuttgart	None	Germany	70174	+49 0711 2842222	None	leonekohle
2	3	François	Tremblay	None	1498 rue Bélanger	Montréal	QC	Canada	H2G 1A7	+1 (514) 721-4711	None	ftremblay
3	4	Bjørn	Hansen	None	Ullevålsveien	Oslo	None	Norway	171	+47 22	None	bjorn.hanse

```
df = pd.read_sql("""
SELECT billing_country, COUNT(*)
FROM invoice
GROUP BY billing_country
ORDER BY COUNT(*) DESC;
""", conn)

df

```

	billing_country	COUNT(*)
0	USA	131
1	Canada	76
2	Brazil	61
3	France	50
4	Germany	41
5	Czech Republic	30
6	Portugal	29
7	United Kingdom	28
8	India	21
9	Ireland	13
10	Chile	13
11	Spain	11
12	Finland	11
13	Sweden	10
14	Poland	10
15	Netherlands	10
16	Hungary	10
17	Denmark	10
18	Australia	10
19	Norway	9
20	Italy	9
21	Austria	9
22	Belgium	7
23	Argentina	5

```
#Q1 – Most senior employee
df = pd.read_sql("""
SELECT *
FROM employee
ORDER BY levels DESC
LIMIT 1;
```

```
""", conn)
df
```

	employee_id	last_name	first_name	title	reports_to	levels	birthdate	hire_date	address	city	state	country	post
0	9	Madan	Mohan	Senior General Manager	None	L7	26-01-1961 00:00	14-01-2016 00:00	1008 Vrinda Ave MT	Edmonton	AB	Canada	

```
#Q2 – Countries with most invoices
df = pd.read_sql("""
SELECT billing_country, COUNT(*) AS invoice_count
FROM invoice
GROUP BY billing_country
ORDER BY invoice_count DESC;
""", conn)
df
```

	billing_country	invoice_count
0	USA	131
1	Canada	76
2	Brazil	61
3	France	50
4	Germany	41
5	Czech Republic	30
6	Portugal	29
7	United Kingdom	28
8	India	21
9	Ireland	13
10	Chile	13
11	Spain	11
12	Finland	11
13	Sweden	10
14	Poland	10
15	Netherlands	10
16	Hungary	10
17	Denmark	10
18	Australia	10
19	Norway	9
20	Italy	9
21	Austria	9
22	Belgium	7
23	Argentina	5

```
# Top 3 invoice total
df = pd.read_sql("""
SELECT invoice_id, total
FROM invoice
ORDER BY total DESC
LIMIT 3;
""", conn)
df
```

	invoice_id	total
0	183	23.76
1	31	19.80
2	92	19.80

```
#City with highest invoice total
df = pd.read_sql("""
SELECT billing_city, SUM(total) AS total_amount
FROM invoice
GROUP BY billing_city
ORDER BY total_amount DESC
LIMIT 1;
""", conn)
df

```

	billing_city	total_amount
0	Prague	273.24

```
#Customer who spent the most
df = pd.read_sql("""
SELECT c.customer_id, c.first_name, c.last_name,
       SUM(i.total) AS total_spent
FROM customer c
JOIN invoice i ON c.customer_id = i.customer_id
GROUP BY c.customer_id, c.first_name, c.last_name
ORDER BY total_spent DESC
LIMIT 1;
""", conn)
df

```

	customer_id	first_name	last_name	total_spent
0	5	František	Wichterlová	144.54

```
#Customers who listen to Rock
df = pd.read_sql("""
SELECT DISTINCT c.email, c.first_name, c.last_name
FROM customer c
JOIN invoice i ON c.customer_id = i.customer_id
JOIN invoice_line il ON i.invoice_id = il.invoice_id
JOIN track t ON il.track_id = t.track_id
JOIN genre g ON t.genre_id = g.genre_id
WHERE g.name = 'Rock';
""", conn)
df

```

	email	first_name	last_name
0	phil.hughes@gmail.com	Phil	Hughes
1	aaronmitchell@yahoo.ca	Aaron	Mitchell
2	terhi.hamalainen@apple.fi	Terhi	Hämäläinen
3	wyatt.girard@yahoo.fr	Wyatt	Girard
4	steve.murray@yahoo.uk	Steve	Murray
5	johavanderberg@yahoo.nl	Johannes	Van der Berg
6	stanisław.wójcik@wp.pl	Stanisław	Wójcik
7	kachase@hotmail.com	Kathy	Chase
8	masampaio@sapo.pt	Madalena	Sampaio
9	jacksmith@microsoft.com	Jack	Smith
10	hholy@gmail.com	Helena	Holý
11	fralston@gmail.com	Frank	Ralston

```
#top 10 Artist
df = pd.read_sql("""
SELECT a.artist_id, a.name AS artist_name,
       COUNT(t.track_id) AS track_count
FROM artist a
JOIN album al ON a.artist_id = al.artist_id
JOIN track t ON al.album_id = t.album_id
JOIN genre g ON t.genre_id = g.genre_id
WHERE g.name = 'Rock'
GROUP BY a.artist_id, a.name
ORDER BY track_count DESC
LIMIT 10;
""", conn)
df

```

	artist_id	artist_name	track_count
0	22	Led Zeppelin	114
1	150	U2	112
2	58	Deep Purple	92
3	90	Iron Maiden	81
4	118	Pearl Jam	54
5	152	Van Halen	52
6	51	Queen	45
7	142	The Rolling Stones	41
8	76	Creedence Clearwater Revival	40
9	52	Kiss	35

```
#track longer then average
df = pd.read_sql("""
SELECT name, milliseconds
FROM track
WHERE milliseconds > (SELECT AVG(milliseconds) FROM track);
""", conn)
df

```

40	mark.taylor@yahoo.au	Mark	Taylor
41	jenniferp@rogers.ca	Jennifer	Peterson
42	jubarnett@gmail.com	Julia	Barnett
43	joakim.johansson@yahoo.se	Joakim	Johansson
44	ellie.sullivan@shaw.ca	Ellie	Sullivan
45	hleacock@gmail.com	Heather	Leacock

					name	milliseconds
46		alero@uol.com.br	Alexandre		Rocha	
0					You Oughta Know (Alternate)	491885
41		roberto.almeida@riotur.gov.br	Master Of Puppets		Roberto	Almeida 436453
2					Snoopy's search-Red baron	456071
50		marc.dubois@hotmail.com			Marc Stratus	Dubois 582086
4					No More Tears	555075
52		emma_jones@hotmail.com			Emma ...	Jones ...
489					Rehab (Hot Chip Remix)	418293
40		fzimmermann@yahoo.de			You Sent Me Flying / Cherry	Zimmermann 409906
491					Amy Amy Amy (Outro)	663426
492		dominiquelefebvre@gmail.com			Symphony No. 3 Op. 36 for Orchestra and Sopran...	Lefebvre 567494
493					Concerto for Violin, Strings and Continuo in G...	493573
58		michelleb@aol.com	Michelle		Brooks	

494 rows × 2 columns

```
#Customer spending per artist
df = pd.read_sql("""
WITH artist_sales AS (
    SELECT il.invoice_id, t.album_id,
           il.unit_price * il.quantity AS amount
    FROM invoice_line il
    JOIN track t ON il.track_id = t.track_id
),
customer_artist_spending AS (
    SELECT i.customer_id,
           al.artist_id,
           SUM(a.amount) AS total_spent
    FROM artist_sales a
    JOIN invoice i ON a.invoice_id = i.invoice_id
    JOIN album al ON a.album_id = al.album_id
    GROUP BY i.customer_id, al.artist_id
)
SELECT cas.customer_id, ar.name AS artist_name, cas.total_spent
FROM customer_artist_spending cas
JOIN artist ar ON cas.artist_id = ar.artist_id
ORDER BY cas.total_spent DESC;
""", conn)
df

```

	customer_id	artist_name	total_spent
0	46	Queen	27.72
1	42	Frank Sinatra	23.76
2	5	Kiss	19.80
3	29	Creedence Clearwater Revival	19.80
4	32	James Brown	19.80
...	...	...	...
2184	33	Itzhak Perlman	0.99
2185	44	Itzhak Perlman	0.99
2186	55	Itzhak Perlman	0.99
2187	8	Emerson String Quartet	0.99
2188	43	Emerson String Quartet	0.99

2189 rows × 3 columns

```
#Most popular genre per country
df = pd.read_sql("""
WITH genre_counts AS (
    SELECT c.country, g.name AS genre,
           COUNT(*) AS purchase_count,
```

```
            ROW_NUMBER() OVER (
                PARTITION BY c.country ORDER BY COUNT(*) DESC
            ) AS rk
    FROM customer c
    JOIN invoice i ON c.customer_id = i.customer_id
    JOIN invoice_line il ON i.invoice_id = il.invoice_id
    JOIN track t ON il.track_id = t.track_id
    JOIN genre g ON t.genre_id = g.genre_id
    GROUP BY c.country, g.name
)
SELECT country, genre, purchase_count
FROM genre_counts
WHERE rk = 1;
""", conn)
df
```

	country	genre	purchase_count
0	Argentina	Alternative & Punk	17
1	Australia	Rock	34
2	Austria	Rock	40
3	Belgium	Rock	26
4	Brazil	Rock	205
5	Canada	Rock	333
6	Chile	Rock	61
7	Czech Republic	Rock	143
8	Denmark	Rock	24
9	Finland	Rock	46
10	France	Rock	211
11	Germany	Rock	194
12	Hungary	Rock	44
13	India	Rock	102
14	Ireland	Rock	72
15	Italy	Rock	35
16	Netherlands	Rock	33
17	Norway	Rock	40
18	Poland	Rock	40
19	Portugal	Rock	108
20	Spain	Rock	46
21	Sweden	Rock	60
22	USA	Rock	561
23	United Kingdom	Rock	166

```
#Highest spending customer per country
df = pd.read_sql("""
WITH spending AS (
    SELECT c.country, c.customer_id,
           SUM(i.total) AS total_spent,
           ROW_NUMBER() OVER (
               PARTITION BY c.country ORDER BY SUM(i.total) DESC
           ) AS rk
    FROM customer c
    JOIN invoice i ON c.customer_id = i.customer_id
    GROUP BY c.country, c.customer_id
)
SELECT country, customer_id, total_spent
FROM spending
WHERE rk = 1
ORDER BY country;
""", conn)
df
```